

Fine-Grained Classification of Beverage Can Products Using Robot Vision

Classification

In this project, I and my groupmates propose a fine-grained image classification task focusing on different beverage can products. All target objects belong to the same super-category of canned beverages, but each category contains some visual differences in logo design, color patterns, font style, and packaging style. Compared to general object classification, fine-grained classification is more challenging because the differences between products are often very small and a bit difficult to discern.

This project will be conducted as a group task where the complete dataset will contain around 20 beverage can classes collected by all group members together including V, Solo, Calm & Stormy, CocaCola Classic, CocaCola Zero Sugar, CocaCola Vanilla, Pepsi, Pepsi Max, Redbull Classic, Redbull Zero Sugar, OxyShred Passion, Fanta Orange, Sprite, Monster Ultra Rosa, Celsius Watermelon Lemonade, Rockstar Tropical Guava, Sunkist Orange, H2 Coco Original, H2 Coco Chocolate, Boss Latte. Each member will be responsible for collecting and managing specific categories. In this proposal, I am going to collect and work on the following four beverage can classes:

1. Kirks
2. V
3. Solo
4. Calm & Stormy

All images for these classes will be collected manually by me using real products purchased from Woolworths. This will ensure the authenticity of the dataset, which has been mentioned in the instruction.

According to me and my groupmates, choosing cans as an object for the project is a feasible option, as cans are easy to collect, easy to bring into the classroom, and can be captured using the robot camera for real-time classification tasks.

Data Collection

For this project, the dataset will be completely created by our group members instead of using any existing dataset from the internet. We are trying to create a more detailed and practical collection pipeline for this project.

Our group will work together to collect around 20 different beverage can classes, while each member will be responsible for specific categories. For my part, I will collect images for four can classes of beverages, for example, Kirks, V, Solo, and Calm & Stormy. I personally purchased all of these cans from Woolworths so that i can create an original dataset and can make sure its reliable.

I am planning to collect all The image in several steps . First, i will collect all the cans and will organize according to their product categories. After that, i will start capturing images using different devices such as smartphone cameras and the classroom robot camera. Once the images are collected, i will try to sorted them into separate folders based on their company name. Finally, we will review all the images and labels carefully to make sure there are no mistakes.

To make the dataset more useful for training, images will be taken in different environments and conditions. For example, photos will be captured from different angles, such as front view, side view, and top view. Images will also be taken from different distances and under different lighting conditions. Some images will be collected indoors with simple backgrounds, while others will be captured outdoors or in more complex environments for example in night at street. Different devices i will use to create variation in image quality and resolution.

In the initial phase, the images will mostly be collected under controlled conditions, such as inside the house and backgrounds should be static, to create a clean dataset for training. Later, additional images will be collected under more challenging conditions like low light, static background with night light, and robot camera movement. This is important because the final system should be able to work properly in real-world environments, not only under perfect situations.

For labeling, each image will be named and organized using the official product name printed on the can. I will try to check manually for every can cheak weather all of th m are in same category or not. This process will help me to improve the overall quality and reliability my dataset.

Importance of the Task

Fine-grained beverage can classification is important in both industry and research because many products look very similar even though they are from different brands or categories. In real-world applications, computer vision systems are already being used in supermarkets, warehouses, and smart retail stores to automatically recognize the products. Accurate product recognition can help to improve management, automated checkout systems, product tracking, and retail robotics.

This task is also useful for robot vision systems where robots need to identify products correctly before performing actions such as sorting or organizing items. Cans are a good example of

fine-grained objects because many cans have similar shapes, colour and layouts, while the differences are often small details such as logo design, text style, or color combinations.

Research by Liu et al. (2016) shows that fine-grained recognition plays an important role in product recognition and retail systems. Similarly, Cui et al. (2018) explain that fine-grained classification tasks are challenging because models must learn very detailed visual differences between similar categories.

Challenges

One of the biggest challenges of this project that might occur is the visual similarity between beverage cans. Many products have almost the same shape, size, and color, which can make classification difficult for this model. Small details such as logo placement, font style, and color patterns become very important for distinguishing between different categories.

Another challenge is variation within the same class. The same can look different depending on lighting conditions, reflections from the surface, camera angle, distance, or background environment. These changes can make the dataset more complex and increase the difficulty for training the model. Research by Wei et al. (2021) highlights that fine-grained classification models need to learn subtle visual features in order to handle these variations successfully.

There are also challenges during real-world deployment using the robot camera. The system needs to work under changing lighting conditions, cluttered backgrounds, motion blur, and different camera qualities. According to Geirhos et al. (2020), deep learning models sometimes focus too much on background information instead of the actual object, which can reduce performance when the environment changes.

Feasibility of the Project

According to me, this project is feasible because cans are easy to find, affordable to purchase, and simple to transport. As I collect all the cans from Woolworths, required images for the dataset will be realistic and manageable. The cans can also be easily brought into the classroom for robot-based testing and image capture.

Images can be collected using personal devices such as smartphones and digital cameras, as well as the classroom robot camera. Since the work is divided among group members, the overall workload becomes more manageable, as each of our members of the group will focus on collecting high-quality images for their assigned classes. Therefore, both the data collection process and the deployment of the classification system can realistically be completed within the given deadline in teamwork.

Sample Images

Including sample images in the final report will help demonstrate the feasibility and diversity of the dataset. Example images of Kirks, V, Solo, and Calm & Stormy cans will be included with different viewing angles, lighting conditions, and backgrounds. These images will show how the dataset has been collected under both controlled and real-world conditions.



Calm and stormy



Kriks



V



Solo

Conclusion

This project proposes a fine-grained classification task involving beverage can products collected manually by group members. My contribution focuses on four beverage can classes: Kirks, V, Solo, and Calm & Stormy. The dataset will be collected using different devices and under different environmental conditions to improve robustness and support real-world deployment.

The project is both technically challenging and practically useful for applications such as retail automation, inventory systems, and robot vision. The planned data collection strategy and phased image collection process will help create a reliable and diverse dataset for training and testing the classification model.

References

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